

Sujin Lee

Postdoctoral Researcher

Safe Transportation Research and Education Center (SafeTREC), University of California, Berkeley

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Research Interests

Travel Demand, Traffic Safety, Urban Informatics, Spatial Analysis, Data Mining, ML/AI

Academic Appointments

Postdoctoral Researcher

Nov. 2024 – *Present*

Safe Transportation Research and Education Center (SafeTREC),
University of California, Berkeley

PI: Julia B. Griswold, Ph.D.

Postdoctoral Researcher

Sep. 2024 – Oct. 2024

Institute of Mechanical Technology,
Korea Advanced Institute of Science and Technology (KAIST)

PI: Kitae Jang, Ph.D.

Education

Ph.D. in Transportation Engineering

Mar. 2019 – Aug. 2024

Graduate School of Mobility, KAIST

Dissertation: Unveiling Functional Regions through Travel Dynamics

Advisor: Kitae Jang, Ph.D.

M.S. in Transportation Engineering

Mar. 2017 – Feb. 2019

Graduate School of Mobility, KAIST

Thesis: Regularity of vehicle trips in urban area

Advisor: Kitae Jang, Ph.D.

B.S. in Urban Planning and Design

Mar. 2012 – Feb. 2017

Hongik University

Computer Engineering (Double major)

Publications

Articles in International Journals

1. Ko, E., **Lee, S.**, Jang, K., & Kim, S. (2024). Changes in inter-city car travel behavior over the course of a year during the COVID-19 pandemic: a decision tree approach. *Cities*, 146, 104758. <https://doi.org/10.1016/j.cities.2023.104758>

2. Yeo, J., **Lee, S.**, Jang, K., & Lee, J. (2023). Real-Time Operations of Autonomous Mobility-on-Demand Services with Inter-and Intra-Zonal Relocation. *IEEE Transactions on Intelligent Vehicles*. <https://doi.org/10.1109/TIV.2023.3299692>
3. **Lee, S.**, Ko, E., Jang, K., & Kim, S. (2023). Understanding individual-level travel behavior changes due to COVID-19: Trip frequency, trip regularity, and trip distance. *Cities*, 135, 104223. <https://doi.org/10.1016/j.cities.2023.104223>
4. **Lee, S.**, Lee, J., Hiemstra-van Mastrigt, S., & Kim, E. (2022). What cities have is how people travel: Conceptualizing a data-mining-driven modal split framework. *Cities*, 131, 103902. <https://doi.org/10.1016/j.cities.2022.103902>
5. Kim, S., **Lee, S.**, Ko, E., Jang, K., & Yeo, J. (2021). Changes in car and bus usage amid the COVID-19 pandemic: Relationship with land use and land price. *Journal of Transport Geography*, 103168. <https://doi.org/10.1016/j.jtrangeo.2021.103168>
6. Shim, J., Yeo, J., **Lee, S.**, Hamdar, S. H., & Jang, K. (2019). Empirical evaluation of influential factors on bifurcation in macroscopic fundamental diagrams. *Transportation Research Part C: Emerging Technologies*, 102, 509-520. <https://doi.org/10.1016/j.trc.2019.03.005>

Articles Submitted

1. **Lee, S.**, Griswold, J., & Lutzker, L. (2026). Multimodal Generative AI for Context-Sensitive Street Classification: An Evaluation across Input Modalities. *Transportation Research Part A: Policy and Practice (under revision)*.
2. **Lee, S.**, Raifman, M., & Griswold, J. (2026). Evaluating the Effectiveness of Automated Speed Enforcement in San Francisco, California: A Quasi-Experimental Approach Using Telematics Data. *Accident Analysis and Prevention (under review)*.
3. **Lee, S.**, Oum, S., & Griswold, J. (2026). Do Telematics Data Replace Crash Data for Road Safety Prioritization? A Comparative Study Across Urban and Rural Contexts. *Journal of Safety Research (under review)*.
4. **Lee, S.**, Griswold, J., & Jang, K. (2026). Identifying functional regions via explainable machine learning on long-term multi-source travel demand. *Transportation Research Part A: Policy and Practice (under review)*
5. **Lee, S.**, Kim, J., Yoon, Y., & Jang, K. (2025). Multi-horizon predictions for parking availability at multiple lots using temporal fusion transformer. *Submitted to Journal of Transport Geography (under review)*

Articles in Domestic Journals (in Korean)

1. **Lee, S.**, Ko, E., Jang, K., Park, S., Park, J., & Yun, I. (2020, October 31). Study on the Adequacy and Improvement of the Threshold Speed of Expressway Congestion. *The Journal of The Korea Institute of Intelligent Transport Systems*.
2. **Lee, S.**, Chai, S., Lee, C. H., An, S., & Jang, K. (2020). Effects of COVID-19 on trip patterns in Daejeon Metropolitan City, South Korea. *The Journal of Korean Society of Transportation*, 17(4), 17-27.

3. **Lee, S.**, Son, S., & Kim, H. (2020). Detour Behavior on the Expressway using Route Travel Data. *The Journal of The Korea Institute of Intelligent Transport Systems*.
4. **Lee, S.**, & Jang, K. (2018). Analysis of Spatial Trip Regularity using Trajectory Data in Urban Areas. *The Journal of The Korea Institute of Intelligent Transport Systems*, 17(6), 96-110.
5. Shim, J., Yeo, J., **Lee, S.**, & Jang, K. (2017). Deriving Macroscopic Fundamental Diagrams Using Probe Vehicle Data Based on DSRC. *The Journal of The Korea Institute of Intelligent Transport Systems*, 16(6), 29-41.

Conferences

Articles in International Conference Proceedings

1. **Lee, S.**, & Jang, K. (2024). Identifying Functional Characteristics of Urban Clusters through Travel Demand Analysis. In *Transportation Research Board Annual Meeting 2024*.
2. Yeo, J., **Lee, S.**, Jang, K., & Lee, J. (2023) Real-Time Operations of Autonomous Mobility-on-Demand Services with Inter- and Intra-Zonal Relocation. In *Transportation Research Board Annual Meeting 2023*.
3. **Lee, S.**, & Jang, K. (2019, October). Regularity of vehicle trips in urban areas. In *2019 IEEE Intelligent Transportation Systems Conference (ITSC)* (pp. 2651-2658). IEEE.
4. Shim, J., Yeo, J., **Lee, S.**, & Jang, K. (2017, July). Effects of Trip Characteristics on Macroscopic Fundamental Diagram in Urban Network. In *Traffic and Granular Flow Conference 2017*. The George Washington University.
5. Shim, J., Yeo, J., **Lee, S.**, & Jang, K. (2017, May). Evaluating Urban Network Performance Using Individual Vehicle Trip Data. In *IEEE International Conference on Mobile Data Management 2017*. KAIST.

Patents and Copyrights

1. Jang, K., **Lee, S.**, Lee, S., Jung, J., & Kim, I. (2024). APPARATUS AND METHOD OF GENERATING A TRAFFIC SIGNAL PLAN USING ON A TRAFFIC CONTROL PLAN, KR Patent No. 10-2024-0181326 (Applied).
2. **Lee, S.** & Jang, K. (2024). Traffic Signal Control Plan Generation Model Using Clustering Techniques. Korea Copyright Commission No. C-2024-039386 (Copyright held by Korea Advanced Institute of Science and Technology (KAIST)).
3. **Lee, S.** & Jang, K. (2023). APPARATUS AND METHOD OF CONTROL TRAFFIC, KR Patent No. 10-2023-0187971 (Applied).

Awards and Honors

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| 2025 | INRIX x MetroLab Challenge Finalist |
| 2025 | PhD Dissertation Award, <i>KAIST College of Engineering</i> |
| 2024 | Start-up Idea Competition 3rd place, <i>Graduate Program for Convergence Technology of D.N.A. Plus</i> |

- 2022 Best paper, In *The 86th Conference of Korean Society of Transportation*
- 2020 Best paper, In *The 82nd Conference of Korean Society of Transportation*
- 2018 Best paper, In *The 2018 Korean Institute of ITS Conference (Spring)*
- 2017 Best poster, In *The 18th IEEE International Conference on Mobile Data Management*
- 2017 Best paper, In *The 2017 Korean Institute of ITS Conference (Spring)*
- 2015 Excellence Scholarship, *Korean Society of Transportation*
- 2014 National Excellence Scholarship (Science and Engineering), *Korea Scholarship Foundation (2014-2015)*

Projects

Current Projects

1. **When Do Stress and Safety Outcomes Tell the Same Story? Evaluating Pedestrian Level of Traffic Stress as a Proxy for Pedestrian Safety Risk at Crossings** 2026–2027
UC Berkeley, SafeTREC
 Role: Lead Researcher, Amount: USD 123,145, Sponsor: U.S. Department of Transportation,
 PI: Maryam Hosseini, Ph.D.
2. **Identifying Harsh Driving Behaviors and Contributing Factors Using Telematics Data: A Case Study in Oakland and Fresno, California** 2025–2026
UC Berkeley, SafeTREC
 Role: Lead Researcher, Amount: USD 113,400, Sponsor: U.S. Department of Transportation,
 PI: Julia B. Griswold, Ph.D.
3. **Evaluate the Effects of California's First Speed Camera Installation Using Telematics Data in San Francisco** 2025–2026
UC Institute of Transportation Studies (UC ITS; funded by California SBI)
 Role: Lead Researcher, Amount: USD 97,471, PI: Julia B. Griswold, Ph.D.
4. **Estimating Wet-Pavement Exposure with Precipitation Data** 2024–2027
UC Berkeley, SafeTREC
 Role: Lead Researcher, Amount: USD 485,000, Sponsor: California Department of Transportation,
 PI: Julia B. Griswold, Ph.D.
5. **Safe System Approach to Setting Speed Limits Phase II** 2024–2026
UC Berkeley, SafeTREC
 Role: Lead Researcher, Amount: USD 350,000, Sponsor: California Department of Transportation,
 PI: Julia B. Griswold, Ph.D.
6. **Pilot Analysis of Telematics Data for Proactive Traffic Safety** 2024–2025
UC Berkeley, SafeTREC
 Role: Lead Researcher, Amount: USD 850,000, Sponsor: Office of Traffic Safety,
 PI: SangHyouk Oum

Selected Completed Projects (during Mater and Ph.D. courses, funded in South Korea)

1. **Deep Learning and GNN-Based Analysis of Metropolitan Mobility Networks** 2021 – 2024

National Research Foundation of Korea

2. **Development of Cloud-Edge based City-Traffic Brain Technology** 2020 –2023
Ministry of Science and ICT (MSIT)
3. **Development of Deep Learning-Based Parking Demand Forecasting Algorithm** 2022 – 2022
Incheon International Airport Corporation
4. **Operation and Management of the Transportation Data Warehouse** 2020 –2021
Daejeon Metropolitan City
5. **TIDY: AI-Driven Big Data Infrastructure for Daejeon City Mobility Services** 2019 – 2021
National Information Society Agency (NIA)
6. **Development of Route-Based Traffic Information Utilization Technologies** 2016 – 2019
Korea Agency for Infrastructure Technology Advancement (KAIA)

Experience

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| Visiting Student | Oct. 2022 – Jan. 2023 |
| University of Virginia (Advisor: B. Brian Park, Ph.D.) | |
| Project: Analysis of route choice behaviors based on trip contexts | |
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Research Intern |
Sep. 2021 – Feb. 2022 |
| Kakao Mobility, AI R&D team (Supervisor: Jungmin Kim, Ph.D.) | |
| Project: Parking demand prediction | |

Talks and Seminars

Invited Talks

1. "Does California's First Speed Camera Installation in San Francisco Reduce Speeding? A Quasi-Experimental Analysis," Webinar, Center for Pedestrian and Bicyclist Safety, Jul 29, 2026 (Virtual; Audience ~200)
2. "Using Replica and Dewey for Traffic Safety Research," SafeTREC Seminar, University of California, Berkeley, Jul 13, 2025 (Hybrid; Audience ~14)
3. "Interactions between People and Cities through the Lens of Transportation Data," SafeTREC Seminar, University of California, Berkeley, Jan 23, 2025 (Hybrid; Audience ~19)
4. "Analysis of Telematics Data and High Injury Network," State of California – Office of Traffic Safety (Technical Assistance), Jul 3, 2025

Guest Lectures

1. "Traffic Safety Analysis Using Telematics Data," University of Seoul, May 6, 2026 (Virtual; Audience ~30)

2. "Traffic Safety Analysis Using Telematics Data," CIVENG C265 001: Traffic Safety and Injury Control, University of California, Berkeley, Apr 16, 2026 (Audience ~25)

Academic Service

Reviewer

CITIES

Journal of Intelligent Transportation Systems: Technology, Planning, and Operations

Journal of Transport Geography

Skills

Proficient in Python, SQL, R, ArcGIS, MS Office Suite (Word, PowerPoint, Excel)

Basic proficiency in Java, Photoshop, Illustrator, AutoCAD, Sketchup, NetLogo

References

Dr. Kitae Jang (Ph.D. Advisor)

Professor, Cho Chun Shik Graduate School of Mobility,

Korea Advanced Institute of Science and Technology (KAIST)

E-mail: kitae.jang@kaist.ac.kr

Dr. Julia Griswold (Postdoctoral Supervisor)

Director, Safe Transportation Research and Education Center (SafeTREC),

University of California, Berkeley

E-mail: juliagris@berkeley.edu

Dr. Euiyoung Kim (Ph.D. Committee Member & Research Collaborator)

Assistant Professor, Faculty of Industrial Design Engineering,

Delft University of Technology (TU Delft)

E-mail: e.y.kim@tudelft.nl

September 1, 2026

Sujin Lee